📰 NewsBot 2.0 — AI News Intelligence System

NewsBot 2.0 is a full end-to-end AI system designed to analyze news articles with the depth and intelligence of a modern newsroom assistant.

It performs classification, topic modeling, sentiment analysis, entity extraction, summarization, semantic search, conversational Q&A, and cross-article insight generation — all inside a clean, interactive Gradio interface.

This project was built as a final course project to demonstrate mastery of machine learning, NLP, system design, evaluation, and user-facing AI application development.

🚀 Features

🔍 **Full Article Analysis**

- News category prediction
- Topic distribution
- Sentiment analysis
- Named-entity recognition
- Entity relationship mapping
- Intelligent summarization
- Multilingual support

💬 **Conversational Q&A**

Ask natural-language questions like:

- “What is the category?”
- “Summarize this article”
- “Which entities are mentioned?”

The system interprets intent and responds using stored article analysis.

🔎 **Semantic Search**

- Encodes the entire BBC dataset into embeddings
- Finds semantically similar articles based on meaning, not keywords

💡 **Insights Generation**

- Extracts common entities
- Aggregates dominant topics
- Computes sentiment distribution
- Produces cross-article intelligence summaries

🧪 **Evaluation & Testing**

Includes:

- Classification metrics
- Topic model coherence
- Summarization quality
- UX evaluation
- Component tests
- Integration tests
- Performance benchmarks

📄 **Interactive Gradio Interface**

Four fully functional tabs:

1. Full Article Analysis
2. Conversational Q&A
3. Semantic Search
4. Insights Generation

🏗 System Architecture

NewsBot 2.0 is composed of eight major engines:

- **AdvancedNewsClassifier**
- **TopicDiscoveryEngine**
- **SentimentEvolutionTracker**
- **EntityRelationshipMapper**
- **IntelligentSummarizer**
- **SemanticSearchEngine**
- **ContentEnhancer** (insight generator)
- **ConversationalInterface** (intent-driven Q&A)

All components are orchestrated by the `NewsBotSystem` master controller.

📚 Dataset

This project uses the **BBC News Classification Dataset**, containing:
- 1,490 labeled training articles
- 735 unlabeled test articles
- Categories: business, tech, politics, sport, entertainment

The dataset is automatically extracted and loaded from Google Drive.

🧠 Model Training

The following models are trained:

- **TF-IDF + Logistic Regression** for news classification
- **LDA Topic Model** for topic discovery
- **VADER** for sentiment analysis
- **SpaCy** for NER
- **Sentence-Transformers** for semantic embeddings

⍰ Testing & Evaluation

The project includes:
- Component-level tests
- Integration tests
- Performance benchmarks
- Summarization quality metrics
- UX evaluation
- A combined evaluation report

🗎 Running the Gradio App

The final interface launches with:

```python
demo.launch(debug=True)
```